

ROSE MARIA

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EXPERIENCE

beataboutmusic

January 2026 – Present

Product Engineer Contributor

Remote

- Architected a rule-based artist recommendation engine using FastAPI and Pandas.
- Analyzed implicit user signals (clicks, likes, follows) via content-based filtering for similar artist discovery.
- Devised a cold start handler using genre-based candidate generation and stratified sampling.
- Developed full-stack features using Node.js, Vue.js, MySQL, and Docker contributing to core platform functionality and deployment workflows.

Crink.app

March 2026 – May 2026

AI Engineer Intern

KSUM, Kochi

- Built a synthetic conversation generation system using GPT-5.4 for 2 LLM-based AI coaches.
- Simulated 30 distinct user personas to stress-test coach response quality.
- Designed an LLM-as-judge pipeline evaluating responses across 6 quality dimensions.
- Benchmarked outputs against 15 therapist-validated ground truths per conversation.
- Built an automated accuracy system, surfacing a 40–50% ground truth pass rate.
- Exposed critical gaps in production AI coach response quality.

Unified Mentor

June 2025 – August 2025

Machine Learning Intern

Remote

- Developed ML solutions spanning regression, classification, and computer vision.
- Built and optimized models using XGBoost, Random Forest, and Logistic Regression.
- Performed feature engineering and hyperparameter tuning across all models.
- Evaluated performance using MAE, RMSE, R^2 , F1-score, and ROC-AUC.

Actionfi Technologies

May 2024

Data Analytics Intern

UL Cyberpark, Calicut

- Performed EDA on a 10,000-row bot performance dataset using Pandas and Matplotlib.
- Generated visualizations and insights into automation pipeline efficiency.
- Built a Gemini API-powered system for structured data extraction from newspaper clippings.
- Developed an LLM benchmarking framework evaluating GPT-3.5 Turbo, GPT-4, and GPT-4o to automate model selection based on comparison against expected outputs.

PROJECTS

Cue – Universal Music Scrobbler 🐙

- A cross-platform music tracking ecosystem that automatically scrobbles playback across 16 Android music applications.
- Built with Flutter and FastAPI, featuring a Gemini 2.5 Flash-powered recommendation engine for lyric, genre, production-credit, and sample-based music discovery.
- Integrated JWT authentication, email OTP verification, and PostgreSQL database storage.
- Used the Spotify Web API for metadata validation and cover art retrieval.

Bylaw Buddy – RAG-Based Legal Chatbot 🐙

- A legal chatbot that answers natural language queries across Indian law PDFs with page-level citations.

- Implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS vector search, and HuggingFace MiniLM embeddings for semantic document retrieval.
- Built PDF highlighting using PyMuPDF and a BrightData web-search fallback.
- Deployed a FastAPI + Streamlit application powered by Groq-hosted Llama 3.3-70B.

Clickflow – Automated Flowchart Digitization

- A 3-tier full-stack platform that converts hand-drawn flowcharts and natural language prompts into editable digital flowcharts.
- Integrated Gemini Flash (multimodal) to transform images and prompts into structured JSON node-edge configurations rendered using ReactFlow.
- Engineered an asynchronous FastAPI backend using SQLAlchemy, Pydantic, and PostgreSQL (JSONB).
- Exposed REST APIs and a custom graph auto-layout engine for interactive editing.

ASL Detection – Real-Time Sign Language Recognition

- A real-time American Sign Language recognition system for webcam-based gesture detection.
- Detected and classified 28 static hand gestures using MediaPipe hand landmarks.
- Generated landmark-based datasets and trained a Random Forest model for live gesture recognition.
- Integrated the trained model with OpenCV for real-time inference on the video stream.

SKILLS

Languages: Python, JavaScript, SQL, C++, Java

AI/ML: LLMs, RAG, Recommendation Systems, Prompt Engineering

Frameworks & Libraries: FastAPI, LangChain, Pandas, Scikit-learn, Flask, Vue.js, React, Node.js

Databases & Cloud: PostgreSQL, MySQL, SQLite, Azure

Tools & APIs: Docker, Git, REST APIs, OpenAI API, Gemini API

ACHIEVEMENTS

Hackathon - TinkHerHack by TinkerHub

Kerala's largest women-only hackathon; recognized as the top team at the venue competition and among the top 300 teams statewide (2-member team). Built Bylaw Buddy, a RAG-based legal chatbot, as the sole backend developer, implementing the RAG pipeline, FastAPI server, LangChain orchestration, and Groq-hosted LLM integration.

EDUCATION

Cochin University of Science and Technology

2023–2027

B.Tech in Computer Science & Engineering

CGPA: 9.52 (S5)

G V H S S for Girls, Nadakkavu

2021–2023

Higher Secondary – Science with Computer Science

99.8%